

Wong Yi Zhen Nicholas

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Summary

Full-stack developer with 3 years of experience building and shipping products in fast-paced environments. Experienced in developing **end-to-end systems**, from CRUD applications to **high-performance, real-time platforms**. Specializes in React, TypeScript, Vue, and WebGL, with a focus on low-latency, data-intensive systems. Leverages AI-assisted workflows to rapidly deliver scalable, production-ready applications.

Education

Monash University Malaysia

Feb 2021 – Nov 2023

Bachelor of Computer Science

- GPA: 3.80/4, CGPA: 3.9/4, WAM: 84/100

Experience

Trontal Group - 2-year startup developing a platform for advanced trading analytics, including multi-exchange data aggregation and real-time market insights

Jun 2025 – Now

Full Stack Developer

- Optimized application load time from ~4s to ~1.5s by analyzing bundle sizes, refining import strategies (eliminating costly barrel imports, reducing unnecessary dependencies), and implementing code splitting. Improved perceived performance through lazy loading and prioritizing critical UI while deferring heavy chart and data rendering.
- Integrated execution trading via a third-party provider connecting to multiple crypto exchanges (Binance, Bybit, Hyperliquid), supporting 100+ users connected through our app. Conducted load testing using k6 to validate system performance and ensure reliability under expected traffic. Collaborated closely with partner engineers to identify and fix bugs, enabling successful delivery within a 1-month timeline.
- Built a real-time crypto price alerting system supporting email and webhook notifications, using WebSockets to push instant in-app alerts to active users and backend APIs to trigger email delivery for offline users. *(60,000+ alerts created since last year August)*
- Core contributor to kScript, a scripting language for trading indicators, with over 140+ scripts written by users. Implemented data fetching and architected flow handling for multiple concurrent data requests, ensuring correct data propagation across asynchronous operations. Designed retry and fallback mechanisms to improve resilience in unreliable network conditions, and built polling strategies to support real-time data updates. Continued maintaining and extending the language with new capabilities.

Dassault Systemes

Dec 2023 – Jun 2025

Full Stack Developer

- Led a frontend team of 3, owning task planning, code reviews, and delivery quality. Partnered with Product Managers to proactively identify risks and ensure stable, predictable releases.
- Increased test coverage from 55% → 95% by developing comprehensive unit tests using React Testing Library and Jasmine; introduced reusable page object patterns to improve end-to-end test maintainability.
- Implemented core features for a logistics visualization platform, enabling users to manage items across stock locations and organize them into planning scopes. Built full-stack functionality using React, TypeScript, Java, and Spring Boot, including REST APIs and access control logic to ensure proper resource ownership and visibility.

Plentisoft Sdn Bhd

Nov 2022 – Feb 2023

Web Developer

- Developed static web pages using HTML, Tailwind CSS, and JS. Stored and displayed data collected with graphs and tables using Express, Angular, Angular Material, and Echarts

Core Skills

- **Languages:** TypeScript, JavaScript, Java, Python
- **Frontend:** React, Vue, Angular, Redux, WebGL, Canvas
- **Backend:** Node.js (Express), Spring Boot, Django
- **Tools & Platforms:** Docker, Git, GitHub, GitLab, Claude Code
- **Cloud:** AWS ([Solution Architect Associate](#)), Google Cloud ([Associate Cloud Engineer](#))

Projects

Portfolio

- Created using NextJS and it took 2 months to design and 1 month of development (excluding working days) to complete it. Inspired from Instagram's User Interface. Comes with Dark Mode and Responsive Design, coded without AI!
- [Portfolio Demo](#)
- [Portfolio Figma](#)

Hackathons

KL Mini Hack

Feb 2023

2nd place

- Participated in a 1-week long blockchain hackathon organized by ETHKL in a team of four and created an anonymous and secure voting system using zero-knowledge proof while incentivizing users with a Cabbage Coin!
- Developed the Cabbage coin in a smart contract and displayed the value on the dashboard using React, Express, Solidity and Truffle
- GitHub Repo [Link](#)

Extracurricular Activities

Dassault Systemes Team Building Committee

September 2024

- Served on HRDC committee, organizing team-building initiatives; sourced and evaluated vendors, negotiated pricing, and coordinated with designers to deliver team merchandise cost-effectively (secured ~RM17/unit pricing).

Monash Board Gaming Club

Feb 2022 – Nov 2022

President

- Lead a team of 12 committees and mentored them on their responsibilities such as event management and sponsorship negotiation
- Collaborated with 10+ board game cafes to provide special deals to our club members
- Improved Club Instagram insights: +3,900% accounts reached, +264% accounts engaged, +34.4% total followers over 2 months

Achievements

Published Paper: Preprocessing Variations for Classification in Smart Manufacturing

Developer/Writer

- Collaborating with my other FYP teammates in university, we managed to publish a paper to the Association for Computing Machinery (ACM). Paper [Link](#).
- Our paper investigates the impact of preprocessing techniques, including brightness and blurriness adjustments, on the performance of deep learning models when dealing with low-quality visual data (edible bird nest) commonly encountered in industrial settings.
- Key Highlights:
 1. Successful development of AI prototype for grading Edible Bird's Nest.
 2. Potential automation for quality control of Edible Bird's Nest.
 3. Utilization of generative AI to create images for training the model.